Universal Fault-Tolerant Computation on Three-Dimensional Layer Codes at Optimal Distance

Research paper

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Abstract

Quantum computers must correct errors, and in three dimensions the layer codes of Williamson and Baspin store quantum information at the highest density that local checks allow. Computing with them requires measuring their logical operators locally, without losing that protection. We give an explicit lattice realization of the layer code for any CSS input code and prove that every pure-X logical operator has a representative on a single lattice layer, at every height, and every pure-Z logical operator one in a single plane, built by a running (prefix) parity along each check. These slabs are measured by gauging with a nearest-neighbor three-dimensional grid of ancillas. We prove that a grid with a constant number of ancillas per edge keeps the full code distance, whereas any measurement layer of bounded thickness cannot, because a thin interface contains a wall of few edges that a few faults can cross. With encoded resource transfer, distillation and a local detector schedule, this gives a universal fault-tolerant architecture under local-stochastic Pauli noise with the optimal distance $\Theta(N^2)$ and $O(Wf + f^{3/2})$ physical qubits for W logical sites. Circuit-level simulations of joint measurements on explicit layer-code blocks (up to 16,293 qubits) confirm the contrast between thin and thick interfaces. On IQM and IBM superconducting processors (6.9 million samples), widening the interface from one edge to two lowers the measurement error 35-fold and 9.7-fold respectively, and repeated rounds protect the reported bit, except where ancilla errors survive a reset, a failure mode outside the noise model.

Keywords: quantum error correction; layer codes; fault-tolerant quantum computation; logical measurement; gauging; superconducting qubits.

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Key terms used in this paper

Qubit	A two-level quantum system; *physical* qubits are the hardware, *logical* qubits the protected information they encode.
Stabilizer (check)	A product of Pauli operators (X, Z) whose measured value is fixed when there is no error; a changed value reveals an error.
CSS code	A code whose checks are all of X type or all of Z type, so that X and Z errors are caught separately.
$[[n, K, d]]$	A code with n physical qubits, K logical qubits and distance d.
Distance d	The smallest number of physical errors that changes the logical information without changing any check.
Fault distance	The smallest number of faults (gate, idle or measurement errors) in a whole circuit that cause an undetected logical error.
Logical operator	An operator that acts on the logical qubits and commutes with every check; its *representatives* differ by checks.
Ancilla	An extra qubit used to measure a check or a logical operator.
Gauging	Measuring a logical operator as the product of many small local checks on an added ancilla grid.
Wall	A set of grid edges whose removal separates the ports of one block from those of the other.
Detector	A parity of measurement outcomes that is fixed without noise; postselection keeps only samples in which every detector is silent.
LDPC	Low-density parity check: every check acts on a bounded number of qubits, and every qubit is in a bounded number of checks.
Θ, O, Ω	"Proportional to", "at most proportional to" and "at least proportional to", for large sizes.

1 Introduction

1.1 Why quantum computers need local error correction

Quantum bits lose their information quickly through noise, so a large quantum computer must correct errors faster than they accumulate. A quantum error-correcting code stores K logical qubits in n physical qubits and repeatedly measures parity checks that reveal errors without disturbing the stored information. Its distance d is the smallest number of physical errors that can change the stored information unnoticed, so a code of large d tolerates many errors.

Real devices also need *geometric locality*: every check and gate should involve only nearby qubits. Locality limits what a code can do. In three dimensions every local stabilizer code obeys $Kd = O(n)$ [1], so $\Theta(d)$ physical qubits per logical qubit is the best possible storage cost. Surface-code patches processed by lattice surgery, today's standard, use $\Theta(d^2)$ [2, 3], and three-dimensional surface and color codes trade larger volume for transversal gates [4]. The layer codes of Williamson and Baspin [5] reach the limit. They embed a CSS code into a local network of surface-code sheets and defects; for balanced good low-density parity-check (LDPC) inputs of length N, the blocks have

$$n = \Theta(N^3), \qquad K = \Theta(N), \qquad d = \Theta(N^2), \quad (1)$$

with bounded-weight, geometrically local checks.

1.2 The problem: storage is not computation

A memory is useful only if it can also compute. For codes of this kind, computation is done by measuring logical Pauli operators, for example the joint parity $\overline{X}_a\overline{X}_b$ of logical qubits in two blocks, through an added system of ancilla qubits [6, 7, 8]. Two obstacles appear for layer codes. A logical operator of a layer code is spread through the whole block, and general measurement schemes for LDPC codes need ancilla graphs with expander properties, which are not local in three dimensions. Moreover, adding measurement layers to a layer code can, in principle, reduce its distance, so the optimal storage could be lost exactly when the code is used.

1.3 Research question, objectives and hypothesis

Research question. Can a three-dimensional layer code perform universal, fault-tolerant quantum computation with nearest-neighbor operations only, while keeping its optimal distance $d = \Theta(N^2)$?

(O1) Build an explicit, checkable lattice for the layer code and find representatives of its logical operators that occupy a single thin layer.

(O2) Design a local logical measurement that keeps the full distance, and identify which measurement interfaces lose it.

(O3) Combine these measurements into a universal fault-tolerant architecture and compute its cost in qubits and time.

(O4) Test the distance mechanisms in circuit-level simulations of explicit layer-code blocks and on superconducting quantum processors.

Hypothesis for (O4). The fault distance of a joint logical measurement is limited by the smallest wall of interface edges between the two blocks. Hence its error should fall sharply when the interface is widened, when parallel ancillas are placed on an interface edge, and when the measurement is repeated over more rounds.

1.4 Background and related work

Logical measurement by gauging, surgery and homological constructions has become a general route to processing quantum LDPC codes [6, 7, 9, 8, 10]. Adapters, extractors and universal architectures address complementary questions of connectivity, parallelism and resource overhead [11, 12, 13, 14]. Local simulations of LDPC codes [15], local fault-tolerance constructions [16], wire codes [17] and planar logical measurements [18] address related geometric questions, while constant-overhead schemes [19, 20, 21] and LDPC magic-state injection [22, 23] rely on nonlocal connectivity. Williamson and Baspin already obtain the layer-code logical operators by quasi-concatenating input logicals [5]. What was missing is a representative that is thin and movable, and a local measurement that provably keeps the distance. Here the organizing constraint is three-dimensional Euclidean locality: logical access, resource delivery and syndrome extraction all use the same nearest-neighbor geometry.

1.5 Contributions of this work

- An explicit lattice realization of the layer code for any CSS input whose checks have weight at least two, with a proof that it encodes exactly the input logical qubits and that every pure-X (pure-Z) logical operator has a representative on a single lattice layer (plane) at every height (position). These *slabs* give the optimal distance $\Theta(N^2)$ for good inputs (Theorem 1).

- A local measurement of the slabs by three-dimensional grid gauging, which keeps the full distance with a constant number of ancillas per edge, and a proof that every measurement layer of bounded thickness loses distance (the *cut obstruction*).

- A universal fault-tolerant architecture using $O(Wf + f^{3/2})$ physical qubits for W logical sites (Theorem 2), with a concrete resource estimate and two structural bounds.

- Circuit-level simulations of a complete joint logical measurement on explicit layer-code blocks, with exact fault distances.

- Demonstrations of the three distance mechanisms on IQM and IBM superconducting processors, which also expose ancilla errors that survive a reset, a failure mode outside the noise model of the theorem.

2 Methodology

2.1 Approach

The work combines three methods. Mathematical proofs give the results for every block size. Computer certificates and circuit-level simulations test the constructions on explicit finite blocks, including exact minimum-fault calculations. Hardware experiments test the distance mechanisms, which already act in circuits small enough for present processors. Figure 1 shows the architecture that the proofs build. The argument has three steps: a prefix-parity construction produces the slab representatives (Sec. 3.1); relative expansion of a cubic grid gives a local gauging measurement that keeps the distance (Sec. 3.2); and repeated extraction, padded composition and encoded resource transfer give a complete circuit (Secs. 3.3 to 3.5).

2.2 The layer code as glued surface-code sheets

A layer code is assembled from surface-code sheets [5]. Each qubit q of the input CSS code becomes a *data sheet*, the plane $y = Y_q$; each input Z check becomes a *Z-check sheet*, a plane $x = \text{const}$; and each input X check becomes an *X-check sheet*, a plane $z = \text{const}$. Every sheet is an ordinary surface code on a rectangle of the cubic lattice, so it carries string operators. A string of X operators that runs from one side of a sheet to the other commutes with all checks of that sheet; a string that stops inside the sheet leaves a violated check, a *defect*, at its end. A Z-check sheet meets the data sheets of the qubits in its check along vertical lines, the *junctions*. Ordered along the check, these are its first, middle and last junctions. The X-check sheets are attached in the same way, with X and Z exchanged.

A junction is a rule that says which strings may end on the line without leaving a defect. At a first or last junction the Z-check sheet ends on the data sheet, and the check-sheet string is present exactly when the data string is. At a middle junction three strings meet the line: the data string, which crosses it, and the check-sheet strings on its two sides. They leave no defect exactly when an even number of the three is present. Writing 1 for a present string, and listing the check-sheet strings first and the data string last,

$$\begin{aligned} H_{\text{end}} &= \begin{bmatrix} 1 & 1 \end{bmatrix}, & \ker H_{\text{end}} &= \{00, 11\}, \\ H_{\text{mid}} &= \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}, & \ker H_{\text{mid}} &= \{000, 110, 101, 011\}. \end{aligned} \tag{2}$$

Williamson and Baspin state these rules as sets of condensed anyons; Eq. (2) is the same information written for X strings. The word 110 is the important one. It lets a string on the check sheet pass a data sheet without any string on that data sheet. If a middle junction allowed only 000 and 111, every check-sheet string would drag a string onto each data sheet it passes, and the lattice would in fact fail to encode the input qubits, as our certificate confirms.

We build every junction explicitly on the lattice. Sheets own distinct qubits, and each junction rule adds one qubit of one sheet to one check of the other. At a middle junction the Z-check-sheet face beside the line gains the data sheet's edge on the line. That face then contains the check-sheet edge on the line, where the string from one side ends; the next check-sheet edge, where the string on the other side begins; and the added data edge. Its check is therefore exactly the middle rule of Eq. (2). The data stars on the line gain the check sheet's edges on the same side, which keeps them commuting with the modified faces. At a first or last junction the check sheet ends on the data line and shares its edges, which gives the two-bit rule.

One more repair is needed. An X-check sheet and a Z-check sheet cross along a line, and there the junction additions of the two sheets make a few stars and faces anticommute, once for each data qubit shared by the two input checks. A row of one-qubit additions along the crossing line, which we call a *lattice wall*, removes these conflicts. It exists exactly when the shared number is even, which is the commutation of the two input checks. The slab operators below never use lattice-wall qubits. Every check lies within unit distance of one lattice vertex and has weight at most six. The junctions have the condensed anyons of Ref. [5]; we do not claim that the lattice is identical to the drawings there, and every result below is proved for this explicit lattice, whose complete check matrices the script `layer_code_explicit.py` in the project repository [24] generates.

2.3 Logical measurement by gauging

Gauging measures a logical operator by measuring many small checks whose product is that operator [6]. Let $\overline{P} = X(D)$ be the selected logical representative. Attach each data qubit of D to its own vertex of a connected grid $G = (V, E)$, call these vertices the *ports* U, and place an ancilla qubit on every edge. The new check at a vertex, its *vertex star*, is X on the edges at that vertex and on the attached data qubit, if any. Every edge belongs to two vertex stars, so the product of all vertex stars is $\overline{P}$, and the product of their measured outcomes is the logical outcome; yet each measurement involves only a few neighboring qubits. The grid also carries Z checks around its unit squares, and every original Z check that meets D is extended along short grid paths joining its ports, so that it commutes with the stars. Preparing the edge qubits in $|0\rangle$, measuring the new checks for R rounds, and finally reading the edges in Z returns the data to its original code with an updated Pauli frame. The X/Z-exchanged construction measures pure-Z logicals.

The danger is that the added qubits create new, shorter logical operators. The quantity that controls this is the expansion of the grid relative to its ports,

$$\beta(G, U) = \min_{S \subseteq V} \frac{|\delta S|}{\min\{|S \cap U|, |U \setminus S|\}}, \tag{3}$$

where δS is the set of edges leaving S and only positive denominators are included. We call δS a *wall*: it separates the ports in S from the others, and β is the smallest number of wall edges paid per port separated.

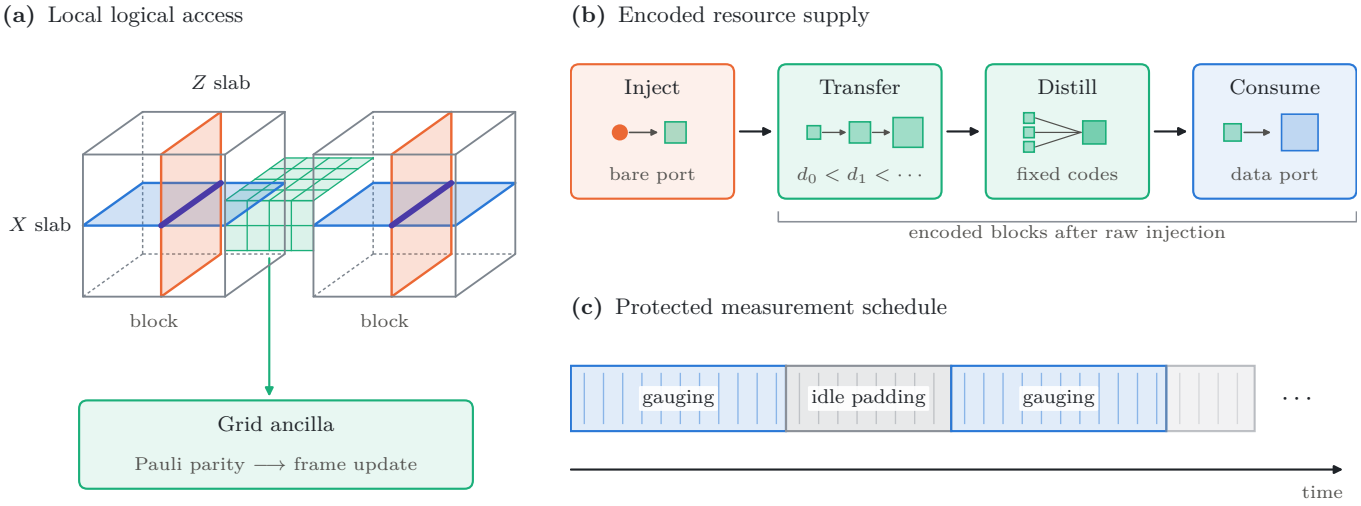


Figure 1: The architecture. (a) A layer-code block exposes single-layer slab ports for pure-X (blue) and pure-Z (orange) logical operators. A local grid ancilla (green) measures the selected Pauli operator; neighboring blocks use the same interface for joint measurements. (b) A raw resource state is injected into an encoded block, transferred through blocks of increasing distance, distilled, and consumed by the data. (c) Repeated extraction and idle padding separate successive measurements. All drawings are schematic; the encoded blocks and grid ancillas occupy three-dimensional regions.

2.4 Noise model and fault distance

We consider local-stochastic Pauli circuit noise: any specified set F_0 of fault mechanisms at distinct locations occurs with probability at most $p^{|F_0|}$; no independence is assumed. A round of check measurements has constant depth and uses only nearest-neighbor gates on a finite ancilla schedule. Each circuit fault acts like at most $c_1 = O(1)$ elementary faults of the simpler phenomenological model, in which a fault is a data error or a wrong measurement outcome; we count faults in that model and divide by c_1 at the end. The *fault distance* of a circuit is the smallest number of faults that flips a protected outcome while leaving every detector silent. For finite circuits we compute it exactly: the script `min_weight_logical.py` builds the detector error model of a circuit with Stim [25] and finds the smallest undetectable logical fault set with the CP-SAT solver of OR-Tools [26].

2.5 Circuit-level simulations

Joint measurement on layer-code blocks. Each block is the explicit realization of the $[[4, 2, 2]]$ input at lattice spacing $s = 2, 3, 4, 5$ ($n = 224$, 504, 896 and 1400 qubits; $d_X = 2s + 1$, $d_Z = 2s$). Logical qubit 1 of two blocks is measured jointly, $\overline{X}_1^{(a)}\overline{X}_1^{(b)}$, by gauging on a grid attached to their single-layer slabs; logical qubit 2 of each block is a spectator. Each logical qubit starts maximally entangled with a noiseless reference qubit, and eight observables test the output state, so the test covers the reported bit, both logical sectors, the spectators and an entangled input. Every check is measured with its own ancilla for $R = 2s + 1$ rounds before, during and after the measurement. The interfaces are a bridge of $w = 1, 2, 3$ grid rows between the blocks, a grid spanning both blocks (one or two ancillas per edge), and, as a baseline, the same blocks left idle.

The noise is uniform circuit noise of strength p from 0.5×10^{-3} to 6×10^{-3}: two-qubit depolarizing noise after every CNOT, one-qubit depolarizing noise on every data and edge qubit in every round, and flips at every reset and measurement, including the final data readout. Each CSS sector is decoded by belief propagation with localized statistics decoding (BP+LSD) [27]. A paired comparison on identical shots shows that this decoder is pessimistic for the grid, by up to a factor of two, and nearly matches the full-model decoder for the bridge, so the contrast reported below is, if anything, understated. Each point ran until 200 failures, a shot cap or the end of a time budget, for 3.0×10^6 shots and 237 core-hours in total, in batches with recorded seeds; error bars are Wilson 95% intervals [28].

Routed circuits. A second set of circuits replaces the per-check ancillas by a routed nearest-neighbor schedule: each ancilla moves by SWAP gates along a spanning tree of reserved routing sites, overlapping checks run in separate color classes, and every waiting qubit suffers idle noise in every time step. It grows the input at $s = 2$, from $[[4, 2, 2]]$ to a six-qubit band code and the Steane code $[[7, 1, 3]]$ ($n = 224$, 720 and 1792; circuits of up to 16,293 qubits), and reports noiseless certification and exact fault distances.

Surface-code gauging campaign. A planar surface code of distance L passes through a base-code → gauged-strip → base-code circuit, without resetting the data between stages, under an SI1000-like model (two-qubit depolarizing strength p, single-qubit and idle strength $p/10$, measurement flips $5p$, reset noise $2p$). The campaign comprises 255 parameter points and 4.534×10^6 shots for $L = 3, 5, 7, 9, 11$ and gauged durations $G = d, 2d, 3d$, decoded by minimum-weight perfect matching [29]. It tests the detector bookkeeping of a code switch, not the layer-code geometry.

2.6 Hardware experiments

Devices. We used the 54-qubit IQM Emerald processor (square lattice) through the Open Quantum platform [30], with qubit rewiring disabled, and the 156-qubit IBM

Heron processors `ibm_fez` and `ibm_kingston` (heavy-hex lattice) through the IBM Quantum Platform with Qiskit [31], using 54 s, 485 s and 407 s of processor time. The data were taken on 13 and 14 October 2026. The blocks are three-qubit repetition codes, not layer codes, so the runs test the distance mechanisms, not the architecture.

Circuits. (i) *Cut obstruction.* Two blocks with checks Z_0Z_1, Z_1Z_2 and $\overline{X} = X_0X_1X_2$ start in $|000\rangle$, and $\overline{X}_a\overline{X}_b$ is measured by gauging on a graph whose bridge between the blocks has $w = 1, 2, 3$ edges. Each vertex star is measured once by its own ancilla, read at the end, so no mid-circuit measurement is needed. The deformed $\overline{Z}_a\overline{Z}_b$ must read $+1$. The detectors are the deformed Z checks and, for $w \geq 2$, the cycle checks of the bridge squares; X on the w bridge edges is a logical operator, so the exact circuit fault distance is w. On IQM the circuits use 16, 18 and 20 CZ gates; routed on `ibm_fez` they need 50, 55 and 60. (ii) *Parallel ancillas.* The single bridge edge carries two ancillas, which adds the check $Z_eZ_{e'}$ and raises the distance to two. (iii) *Repeated rounds.* Two phase-flip blocks ($d_Z = 3$) joined by one edge measure their two vertex stars and four block checks in R rounds; the reported bit is the product of the stars, and its exact circuit fault distance is $\min\{R, 3\}$. On IQM, which does not allow a measured qubit to be reused, every round uses fresh ancillas ($R = 1, 2$); on `ibm_kingston` the ancillas are measured and reset between rounds ($R = 1, 2, 3$; run B also tests two resets between rounds).

Placement and checks. Apart from the routed `ibm_fez` circuits, every two-qubit gate acts on a physical coupler, so the circuits contain no SWAP gate; the IQM placements are SWAP-free under both possible qubit numberings of the platform. On IBM the repeated-rounds layout was chosen from the live calibration. Before submission each circuit as sent was checked to equal the reference circuit, to pass every detector without noise, and to have the stated exact circuit fault distance.

Analysis. Each result is the error of the observable after postselection on every detector, with Wilson 95% intervals; comparisons use one-sided Fisher exact tests [32]. As a control, the same samples are reanalyzed with the checks that the better interface adds ignored. For the later runs the test and its level 0.05 were fixed in the analysis script before the jobs were submitted; for the first runs the analysis code was fixed before the data, but the tests were chosen afterwards. The bit order of each output was verified from the data wherever possible. Predictions come from Pauli noise models with the devices' error rates. Two attempts on a Rigetti processor were excluded because most of their detectors fired at random (in the second, one qubit read 1 in all 30,000 shots although it was listed as working); their outputs are archived.

3 Results

3.1 Logical operators in single-layer slabs

Theorem 1 (Uniform slab representatives)**.** *For every input CSS code with checks of weight at least two, the explicit realization encodes exactly K logical qubits. Every pure-X logical class has a representative supported on the single layer of z-edges spanning $[z_0, z_0{+}1]$, for every height z_0 of the block, and every pure-Z class has a representative in the plane $x = x_0$, for every x_0. The heights and planes are chosen independently of the logical classes.*

Construction. The operator is built at one fixed height z_0, one input check at a time. Choose a pure-X logical $x \in \ker H_Z$ of the input code and put an X string at height z_0 on the data sheet of every qubit with $x_q = 1$. Each string crosses the junction of every Z-check sheet attached to its data sheet and, alone, would leave a defect there. Along one Z-check sheet the selected junctions must therefore be joined in pairs by strings on the check sheet at the same height, which we call the *backbone*. Pairing is possible because x commutes with the input check, so the check contains an even number of selected qubits. Walk along the check from its first qubit and switch the backbone at every selected junction. With ordered incidences $x_1, \ldots, x_k$, the backbone after the ith junction is present exactly when an odd number of selected qubits lies behind it, a running count that we call the *prefix parity*,

$$p_0 = 0, \qquad p_i = \sum_{h=1}^{i} x_h, \qquad p_k = 0, \tag{4}$$

where $p_k = 0$ is the commutation $\sum_i x_i = 0$ over $\mathbb{F}_2$. At every middle junction $p_{i-1} + p_i + x_i = 0$, so none or two of the three strings are present, as Eq. (2) requires; at the ends $p_1 = x_1$ and $p_{k-1} = x_k$. No defect remains anywhere (Fig. 2). Every piece lies in the same layer of vertical edges, so the operator is one lattice layer thick, whatever the block size and whichever logical is chosen. Building it at another height gives another representative of the same class. In the explicit realization the check is exact and local: every stabilizer meets the operator in one of the forms $x_1 + p_1$, $p_{i-1} + p_i + x_i$, $p_{k-1} + x_k$ of a single incidence, or not at all.

Exactly K logical qubits. The pure-Z operators are built in the same way in a vertical plane $x = x_0$, with backbones on the X-check sheets. An X slab operator and a Z slab operator share only one data edge for each input qubit selected by both, so they anticommute exactly when the two input logicals do. This gives K independent logical qubits. A count shows that there are no others. On its own, each data sheet would encode one qubit and each check sheet would remove one; the junctions and lattice walls add neither qubits nor checks; and every redundancy among the lattice checks comes from a redundancy among the input checks. Hence the realization encodes exactly K qubits, and the slab operators represent every pure-type class. A certificate on complete codes with up to 10,936 qubits, including the linear forms above and exhaustive negative controls, checks each step.

Distance. The same ports give the distance, by a simple idea: a logical operator must be visible in every plane that contains a complete port of the other type. Let O be a nontrivial X logical. It anticommutes with some Z logical, which has a slab representative in each of the $L_x + 1$ disjoint planes $x = x_0$, so O must place qubits in every one of these planes. Cut the Z slabs of one plane into pieces, a data column for each input qubit

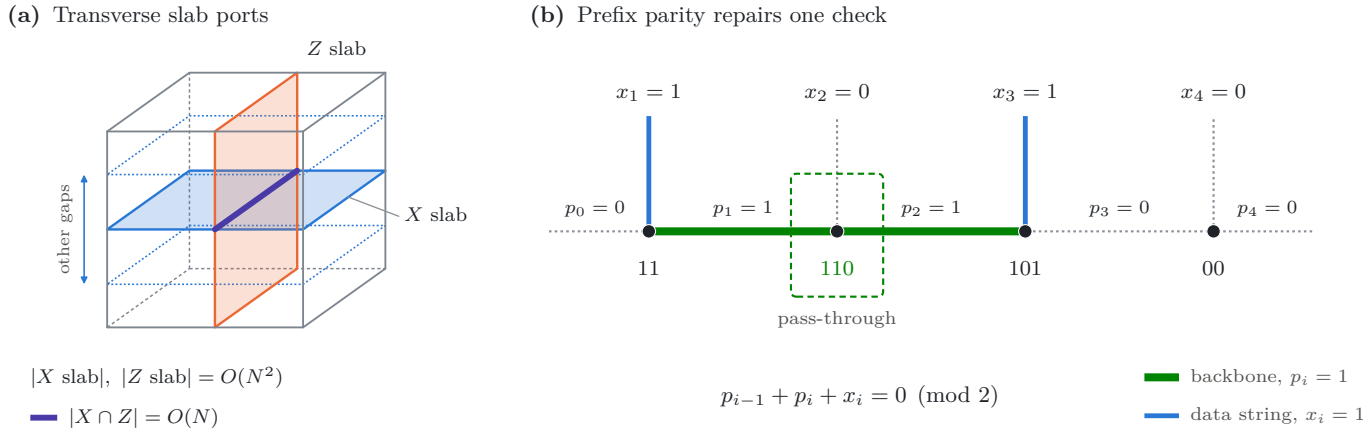


Figure 2: Slab representatives and the prefix parity. (a) Complete X and Z ports occupy single-layer slabs. Each slab contains $O(N^2)$ qubits, and two crossing slabs share $O(N)$. Translating a slab to another height (or plane) gives another representative of the same logical class. (b) For the incidence word $(1,0,1,0)$ the running parity is $(0,1,1,0,0)$. The highlighted middle cell uses the word 110: the backbone stays on although the data string is absent. End cells use the two-bit rule of Eq. (2).

and the backbone segments, and record which pieces O meets an odd number of times. Read as input qubits, the records spell out a nontrivial X logical of the input code, with at least d_X^{in} entries. A backbone piece stands for at most $\eta_X = \max\{1, \lfloor w_X/2 \rfloor\}$ input qubits, where w_X is the largest input X-check weight, because the first t qubits of a check may be exchanged for the rest. Hence

$$d_X \geq (L_x + 1) \left\lceil \frac{d_X^{\text{in}}}{\eta_X} \right\rceil , \tag{5}$$

dually for Z. For balanced good inputs the realization therefore has $n = \Theta(N^3)$, $K = \Theta(N)$ and $d = \Theta(N^2)$, the optimal parameters of Eq. (1). For the $[[4,2,2]]$ and $[[6,4,2]]$ inputs the bound equals the exact distance in every computed instance. For balanced inputs the block has $\Theta(N)$ heights and $\Theta(N)$ transverse positions, so there are $r_X, r_Z = \Theta(N)$ pairwise-disjoint slab regions of each orientation, each a complete port.

What is new relative to Ref. [5] is an explicit realization in which this construction can be verified check by check, a single-layer thickness independent of the block size and of the chosen logical, and its uniformity over every height, which supplies the disjoint ports used below.

3.2 A three-dimensional grid keeps the distance

Deformed distance. For a code with CSS distances d_X, d_Z, the deformed distances satisfy

$$\widetilde{d}_X \geq \min\{1, \beta(G,U)\} d_X, \qquad \widetilde{d}_Z \geq d_Z. \tag{6}$$

The reason is short. An X logical of the deformed code may put part of its weight on the ancilla edges, but those edges must form a wall δS; any other edge set anticommutes with some square Z check. Multiplying by the vertex stars in S removes the wall and puts X instead on the data qubits attached in S, or, using the complement of S, on those attached outside it. Either way the result is a logical of the original code, so it has weight at least d_X. The trade removes $|\delta S|$ edges and adds at most $m = \min\{|S \cap U|, |U \setminus S|\}$ data qubits, while $|\delta S| \geq \beta m$. Hence the deformed logical weighs at least $\min\{1, \beta\}$ times a logical of the original code. The Z sector cannot use the ancilla at all: a Z logical reduces to an original Z logical times grid cycles.

Why three dimensions. A slab representative lies in one plane and has $\Theta(N^2)$ ports. If the grid were only that plane, a straight line across it would wall off half of the ports with $\Theta(N)$ edges, so $\beta = O(1/N)$. If the grid is a box as thick as it is wide, a wall must be a surface. Either it separates roughly the same region in every layer of the box, and then it cuts through all the layers, or the separated region changes from layer to layer, and every change costs edges between layers. In both cases its area grows in proportion to the number of ports it separates. If U occupies at most t parallel planes of a box with aspect ratio at most A, this gives

$$\beta(G,U) \geq \frac{1}{2At}. \tag{7}$$

Constant refinement assigns distinct ports to distinct grid vertices and keeps the check weight and incidence constant. Thus one block, or two blocks at fixed separation, admit local logical measurement using $O(n)$ ancillas and retaining distance at least $\beta_0 d$ for a fixed $\beta_0 > 0$.

The cut obstruction. A logical measurement can lower the fault distance in three ways: the added qubits can carry a short logical operator; a few wrong outcomes can flip the reported result; and a few faults can exploit the join between two consecutive operations. The first danger is real for thin interfaces. Once $\overline{X}_a \overline{X}_b$ has been measured, it is a stabilizer, so $\overline{X}_a$ and $\overline{X}_b$ act in the same way on the code. Take any wall δT with the ports of block a on one side and those of block b on the other. The product of the vertex stars in T is $\overline{X}_a$ times X on the wall, so $X(\delta T)$ alone is a logical operator equivalent to $\overline{X}_a$ [Fig. 3(a)], and

$$\widetilde{d}_X \leq \min\{|\delta T| : \ T \cap U = U_a\}. \tag{8}$$

For layer-code blocks of side $\Theta(N)$, the distance $d = \Theta(N^2)$ is comparable to the area of a block face. A measurement layer of bounded thickness h meets the plane between the blocks in a strip of $O(hN)$ edges, which is a wall, so it would reduce the distance of every joint

measurement to $O(hN)$. Joint measurements cannot be avoided: they implement CNOT, phase teleportation and resource transfer. Computer checks on explicit codes confirm the loss: a single added sheet leaves distance at most L where the undeformed code has L^2.

Four ingredients keep the distance. First, the grid fills the participating blocks, so every wall between their ports is a surface with $\Omega(N^2)$ edges [Fig. 3(b)]; this is the constant relative expansion of Eq. (7), which the constant slab thickness makes possible. Second, carrying $m_0 = \lceil 2At \rceil$ parallel ancillas on every grid edge multiplies every wall by m_0, so $\beta \geq 1$ and Eq. (6) gives $\widetilde{d}_X \geq d_X$: each measurement keeps the full static distance at constant ancilla cost. Third, the measurement is repeated in time [Fig. 3(c)]. A single wrong outcome of one vertex star flips the reported product, so the new checks are measured for $R = \Theta(d)$ rounds, and a wrong outcome that is not repeated is caught by comparing consecutive rounds. Williamson and Yoder proved the resulting bound for gauging [6]; because our operations start and end in open idle rounds rather than at perfect boundaries, we give a direct proof. Faults of the measured type (X) leave an error on the data and a set of flipped edge readouts that together form an operator of the deformed code, which is a nontrivial logical whenever they cause an error, so there are at least $\widetilde{d}_X$ of them. Faults of the other type either leave a logical error on the data, with at least d_Z faults, or flip the reported outcome; then either every one of the R rounds contains a wrong outcome, or a round without one shows that the errors so far form a logical that anticommutes with the measured one, again with at least d_Z faults. Hence the measurement has fault distance at least $\min\{\widetilde{d}_X, d_Z, R\}$, and exact minimization on small instances shows that this bound is attained. Fourth, successive measurements on a block are separated by $\Theta(d)$ idle rounds, so faults that straddle two operations also cost $\Omega(d)$ (Sec. 3.4).

Resource transfer is the one place where blocks of different distance meet. Joining a small block to a large one cannot expose the large block: the same wall argument, applied to everything except the small block's own logical qubits, shows that every other logical operator keeps the distance of the larger block. The small courier blocks below therefore carry only one traveling resource each, and their errors are charged to distillation. Every logical qubit that is read later, other than a resource in transit, is protected at every step by fault distance $\Omega(d)$.

3.3 Gates and encoded resources

Pure-X/Z measurements suffice for the logical gate set. A CNOT uses an ancilla a in $|+\rangle$ and the measurements

$$Z_c Z_a \ (u), \qquad X_a X_t \ (v), \qquad Z_a \ (w), \tag{9}$$

followed by the Pauli-frame update $Z_c^v X_t^{u \oplus w}$, where outcome bit zero denotes eigenvalue $+1$. For phase teleportation, prepare a in $R_\theta|+\rangle$ with $R_\theta = \mathrm{diag}(1, e^{i\theta})$, measure $Z_d Z_a$ with bit m and X_d with bit n, and retain a. The correction

$$C_{mn} = R_{2\theta}^m Z_a^n X_a^m \tag{10}$$

produces $R_\theta|\psi\rangle$. For $\theta = \pi/2$ this consumes $|Y\rangle = S|+\rangle$; for $\theta = \pi/4$ it consumes $|A\rangle = T|+\rangle$, with a possible S correction using one additional $|Y\rangle$. Exchanging X and Z gives $S_X = HSH$, and $SS_XS = e^{i\pi/4}H$. These identities generate the universal Clifford+T set from one- and two-block measurements.

A physical resource enters at a designated bare port. The raw qubit is teleported into an encoded block by a joint ZZ measurement and an X readout; all spectator logical classes keep distance $\Omega(d)$ in the quotient by the resource's weight-one error class. Subsequent transfers use only encoded blocks, with distances growing by fixed factors; we call this sequence an encoded courier. The resource supply uses fixed Steane $7 \to 1$ and punctured Reed–Muller $15 \to 1$ factories built from the same logical primitives [33, 34, 35]. Resource errors are modeled as local-stochastic defective slots: a defective candidate may be in any state, entangled with other candidates or with the environment. Phase teleportation with any such resource applies, relative to the ideal rotation, an operator in $\mathrm{span}\{I, Z\}$ on the coordinate it feeds, so a predetermined recovery returns exactly the ideal output whenever at most one coordinate of a run is defective. Each run therefore always produces an output, and the outputs are again a defective family with

$$q_{r+1} \leq C_{\mathrm{dist}} q_r^2, \qquad C_{\mathrm{dist}} = \tbinom{30}{2} = 435, \tag{11}$$

counting pairs among the fifteen principal and fifteen correction slots of a run. No twirl or diagonal error basis is assumed, and the raw boundary, encoded transfers and factories are included in the circuit analysis rather than supplied as ideal resources.

3.4 Fault tolerance of the complete circuit

The argument that the whole computation keeps fault distance $\Omega(d)$ has a simple shape. An undetected logical error needs a fault pattern that no detector notices. Split such a pattern into connected pieces, where two faults are connected when they can trigger a common detector; at least one piece causes the error. A piece inside one operation and the idle rounds around it needs at least $\min\{d, R\}$ faults, by Sec. 3.2. Successive operations on a block are separated by R_{pad} idle rounds, and two connected faults are at most $\tau = O(1)$ rounds apart, so a piece that reaches from one operation to the next must stretch across the padding. Precisely, with δ_g^{rel} the fault distance of measurement window g and $\delta_{\mathrm{mem}}^{\mathrm{rel}}$ that of idle memory, the composed phenomenological distance satisfies

$$f_{\mathrm{ph}} \geq \min\left\{ \min_g \delta_g^{\mathrm{rel}}, \delta_{\mathrm{mem}}^{\mathrm{rel}}, 1 + \left\lfloor \frac{R_{\mathrm{pad}} - \tau}{\tau} \right\rfloor \right\}, \tag{12}$$

with $\delta_g^{\mathrm{rel}} \geq \min\{\beta_0 d, R_g\}$ for a window of R_g rounds and $\delta_{\mathrm{mem}}^{\mathrm{rel}} \geq d$. Choosing R_g and R_{pad} of order d makes every term $\Omega(d)$.

The technical work lies in making "inside one operation" precise without assuming that each operation starts from a perfect state. We cut the schedule in time at idle rounds and carry three pieces of data across each cut: the syndrome s left by earlier errors, the logical Pauli

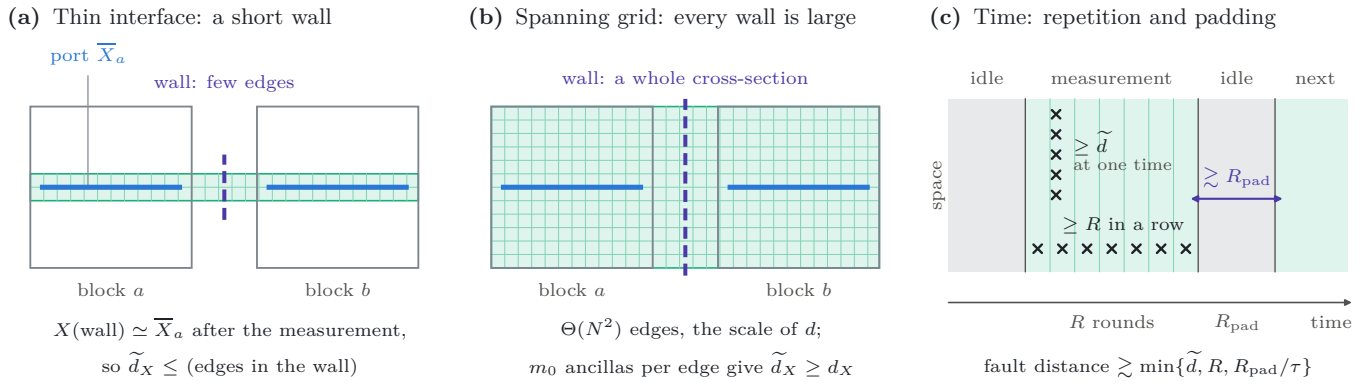


Figure 3: Why the measurement keeps the fault distance, in a side view of two blocks (squares) with their X slab ports (blue) and the ancilla grid (green). (a) If the grid joins the blocks only through a thin bridge, X on a wall across the bridge (dashed) is equivalent to $\overline{X}_a$ once $\overline{X}_a\overline{X}_b$ is measured, so a few faults cause a logical error. (b) If the grid fills both blocks, every wall between their ports is a whole cross-section, with $\Theta(N^2)$ edges, the same scale as the code distance; m_0 ancillas per edge make every wall at least as heavy as the ports it separates, and the full distance is kept. (c) In time, an undetected error inside a measurement is either a logical at one time step, with at least $\widetilde{d}$ faults, or one outcome wrong in each of the R repeated rounds. Idle rounds between operations force any pattern that links two of them to stretch across the padding.

correction λ accumulated so far, and the outcomes ρ of the last measurement round, which the next detectors compare against. The records must be carried separately, because a wrong outcome in the last round before a cut changes ρ but not the state. With these three variables adjacent operations compose exactly; a direct simulation of an idle $\to$ gauging $\to$ idle $\to$ gauging $\to$ idle schedule on explicit layer-code blocks agrees with the composed map on every single fault, and gluing without ρ fails. Each piece of a fault pattern is then judged by its effect relative to the same schedule without it, which removes every dependence on earlier errors.

Resource errors need one more step, because the courier blocks have smaller distance. A new error first created at courier stage j needs a connected fault pattern of size $\Omega(d_j)$; the raw injection stage is charged separately, with a constant. Distinct resources are kept far apart in spacetime, so one short connected pattern cannot corrupt two of them: a chain of connected faults can leave the courier blocks of one resource only through its transfer into the factory block, and from there it must cross the R idle rounds that separate the transfer from every other operation on that block. A script builds a complete factory schedule by these rules, with cell reuse and adaptive branches, and checks this separation on it. Summing over the stages gives a resource error $q_0 = O(\sqrt{p})$, which Eq. (11) then suppresses quadratically.

Every detector has bounded support, so the graph that connects faults through common detectors has bounded degree. The committed decoder chooses a minimum-cardinality fault set consistent with the detector history through a fixed buffer beyond the commitment time. A wrong decision about a protected outcome requires a large connected set of fault locations, a fixed fraction of which are actual faults, and counting such sets in a bounded-degree graph yields a positive threshold. This decoder is information-theoretic; efficient classical decoding is not assumed.

3.5 Architecture theorem and resource scaling

The input family is a balanced good CSS LDPC family with boundedly dense sizes: every sufficiently large target length has an available member within a fixed factor. Fixed-degree quantum Tanner codes provide such a family [36, 37]. Let $A \subset \mathbb{Z}^3$ be the logical region, $W = |A|$, and T the logical circuit depth. Partition the logical lattice into boxes of side $b(f) = \Theta(f^{1/6})$, and let $B_f(A)$ count the boxes that meet A but are not completely filled.

Theorem 2 (Universal fault-tolerant architecture)**.** *Use the explicit realization of Theorem 1, a balanced good input family with boundedly dense sizes, and the local measurement, resource and extraction schedules above. There are constants $p_{\text{th}}, c, f_{\min} > 0$ such that for $0 < p < p_{\text{th}}$ and target failure $0 < \varepsilon < 1/2$, any finite nearest-neighbor three-dimensional logical circuit can be simulated with*

$$f = \max\left\{ f_{\min}, \left\lceil \frac{c \log(WT/\varepsilon)}{\log(p_{\text{th}}/p)} \right\rceil \right\}, \tag{13}$$

using

$$Q_{\text{phys}} = O\Big(Wf + (B_f(A) + 1) f^{3/2} \Big), \tag{14}$$

$$D_{\text{phys}} = O(T f^{3/2} \Psi), \qquad \Psi = O(f^{\log_2 15}). \tag{15}$$

Persistent data remain in layer-code blocks, and physical operations use nearest-neighbor connectivity. For a connected bounded-aspect logical box, $Q_{\text{phys}} = O(Wf + f^{3/2})$.

The scaling follows directly from the code parameters. Choose $d = \Theta(f)$ and measurement, padding and commitment buffers of order f. Equation (1) then gives $N = \Theta(\sqrt{f})$, $K = \Theta(\sqrt{f})$ and $n = \Theta(f^{3/2})$ per cell. A full coarse box fits into one cell, so $O(W/K + B_f(A) + 1)$ cells suffice, and multiplication by n proves Eq. (14); for a bounded-aspect box $B_f(A) = O((W/\sqrt{f})^{2/3} + 1)$ is absorbed (Fig. 4). For time, a distilled resource takes $O(f\Psi)$ rounds, where ℓ distillation levels give $\Psi = O(15^\ell)$ and $2^\ell = O(f)$. One logical layer needs $O(K)$ resources per data block, and serial production gives $O(Kf\Psi) = O(f^{3/2+\log_2 15}) \subseteq O(f^{5.41})$ rounds per

layer, against $O(f)$ for lattice surgery. The serial schedule minimizes space, not space-time volume: with P factories per cell the space becomes $O(WfP+f^{3/2})$ and the depth $O(TKf\Psi/P)$.

Concrete estimate. An estimate with stated constants, explicit integer allocations and distances chosen from each schedule's complete runtime makes the scales explicit. For $W = 100$ sites, $T = 10^6$ layers, $\varepsilon = 10^{-2}$, $p = 10^{-3}$ and injected resource error 10^{-2}, the always-output distillation covered by the theorem does not converge: Eq. (11) requires an injected error below $1/C_{\mathrm{dist}} \approx 2.3 \times 10^{-3}$. With optimistic postselected distillation, which the theorem does not cover, the serial layer-code schedule needs about 10^9 qubits for two months, against 4×10^5 qubits for three days for a serial surface-code layout and 2×10^8 qubits for 23 s for a throughput-limited one. At these distances a layer-code block holds one to three logical qubits; with the same accounting on both sides it stores a logical qubit more cheaply than a surface-code patch only for $d \gtrsim 1100$ (between about 600 and 2300 across scenarios). The asymptotic advantage is therefore far from the practical regime until these constants, and the distillation protocol, improve.

3.6 Port packing and shallow logical gates

Slab mobility also constrains how logical information can be spatially accessible. For a region A, let $g_X(A)$ be the dimension of the pure-X logical space supported in A, and define $g_Z(B)$ analogously. The rank $r(A,B)$ of their commutation pairing obeys

$$\max\{0, g_X(A) + g_Z(B) - K\} \le r(A,B) \le |A \cap B|, \quad (16)$$

because all cross commutation factors through the intersection and the full logical pairing is nondegenerate. If $A_1, \dots, A_{r_X}$ are disjoint complete X ports and $B_1, \dots, B_{r_Z}$ disjoint complete Z ports, summing over all pairs gives the crossbar bound

$$r_X r_Z K \le n. \quad (17)$$

For the balanced layer-code slabs $r_X, r_Z = \Theta(N)$, $K = \Theta(N)$ and $n = \Theta(N^3)$, so the bound is saturated: the cubic size is optimal among CSS families required to provide $\Theta(K)$ disjoint complete ports in both polarizations. Pairing the ith X-slab and Z-slab representatives of any logical Pauli also gives disjointness $\Delta = \Theta(N)$, with smallest and largest logical weights $\Theta(N^2)$. The Jochym-O'Connor–Kubica–Yoder criterion [38] then forces every fixed-depth, bounded-arity, code-preserving unitary on a large static block to be logical Clifford. This is why the architecture uses adaptive measurements and resource states, which lie outside that unitary setting.

3.7 Simulations on explicit layer-code blocks

Exact fault distances. Without noise every detector and all eight observables are deterministic, so the circuit implements the projective measurement with the correct outcome and leaves the spectators untouched. On two $[[4,2,2]]$ blocks at $s = 2$ ($n = 224$, distances 5 and 4), the fault distance of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ through a bridge of $w = 1, 2, 3$ rows is exactly w; for $w = 1$ a single fault suffices [Fig. 5(b)]. This is the cut obstruction at circuit level. A grid spanning both blocks restores it to exactly 5, the blocks' X distance, while the fault distances of the reported bit, set by the number of rounds, and of the spectators do not depend on the interface. At $s = 3$ the bridge distance is still exactly w, although the blocks' X distance is now 7.

Failure rates. At $p = 10^{-3}$ the failure rate of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ through the one-row bridge rises from 3.9% at $s = 2$ to 8.8% at $s = 5$, because the undetectable bridge fault has more rounds in which to occur, while through the spanning grid it falls from 8.2×10^{-4} to about 10^{-4} and agrees with the same blocks left idle [Fig. 5(a)]. For the spanning grid each unit of spacing suppresses failures by a factor $\Lambda = 4.8$ in the X experiment and 2.6 in the Z experiment at $p = 10^{-3}$, falling to one near $p \approx 6 \times 10^{-3}$ [Fig. 5(c)].

Routed circuits. Without noise all 28 routed measurement circuits implement the measurement exactly, and the 12 idle controls also give silent detectors and correct observables. Through the one-row bridge the fault distance is exactly 1 for all three inputs, although the blocks' X distance grows from 5 to at least 18. On the $[[4,2,2]]$ blocks with $R = 5$ the spanning grid gives exactly 5, the same as the blocks left idle; the reported bit has exactly $5 = R$, and the spectator frames exactly $4 = d_Z$, as in idle memory, so routing leaves the distances of the colocated circuit unchanged where both are resolved. On the band blocks the reported bit has an undetectable pattern of weight 6 at $R = 7$, which the block itself explains: its lightest Z-type operator that anticommutes with the measured logical has weight 6. At the tested noise strengths the sampled failure rates of these long routed circuits do not distinguish the interfaces, so we use only their exact distances.

Surface-code campaign. All 15 noiseless controls have zero failures. For every gauged duration the failure rate decreases monotonically with L across all five sizes at every tested $p \le 0.004$. At $p = 0.0035$ increasing L from 3 to 11 reduces the failure rate from 0.1111 to 0.0324 for $G = d$, and at $p = 0.0015$ (also $G = d$) the reduction is about 6.5×10^2. Tripling the gauged duration shifts the smallest-versus-largest crossing by about 3.2%, from 4.47×10^{-3} to 4.33×10^{-3}. These are finite-size diagnostics of a surface-code model, not estimates of the theorem's threshold.

3.8 Hardware demonstrations

Figure 6 shows what was run and what was measured, and Table 1 collects the tests. Each of the three predicted gains appears on hardware; the exceptions are noted below.

Cut obstruction. The postselected error falls 35-fold, from 2.4% at $w = 1$ to 0.070% at $w = 2$, on IQM (pooled over three runs on two sets of qubits; $p \le 9.5 \times 10^{-19}$ in each of the two runs with both widths), and on IBM from 5.8% to 0.59% and 0.14% at $w = 1, 2, 3$, by factors of 9.7 and 4.3 [Fig. 6(a,d)]. When the cycle checks that the extra bridge edges create are ignored, the error returns to the width-one level (2.2 to 3.3% on IQM, 7.2 to 7.9%

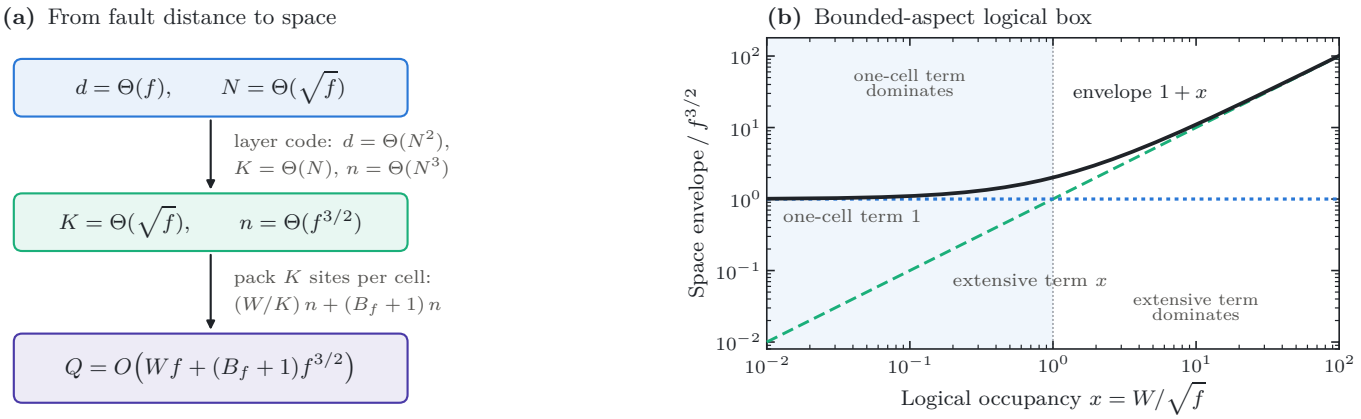


Figure 4: Origin and crossover of the resource bound. (a) Target fault distance f fixes the cell parameters; packing $K = \Theta(\sqrt{f})$ logical sites per cell yields an extensive Wf term plus the cost of partially occupied cells. (b) For bounded-aspect boxes, the normalized scaling envelope is proportional to $1 + x$, where $x = W/\sqrt{f}$. The one-cell term dominates for $x \ll 1$ and the extensive term for $x \gg 1$. Constants are suppressed: this illustrates the asymptotic bound and is not a numerical qubit estimate.

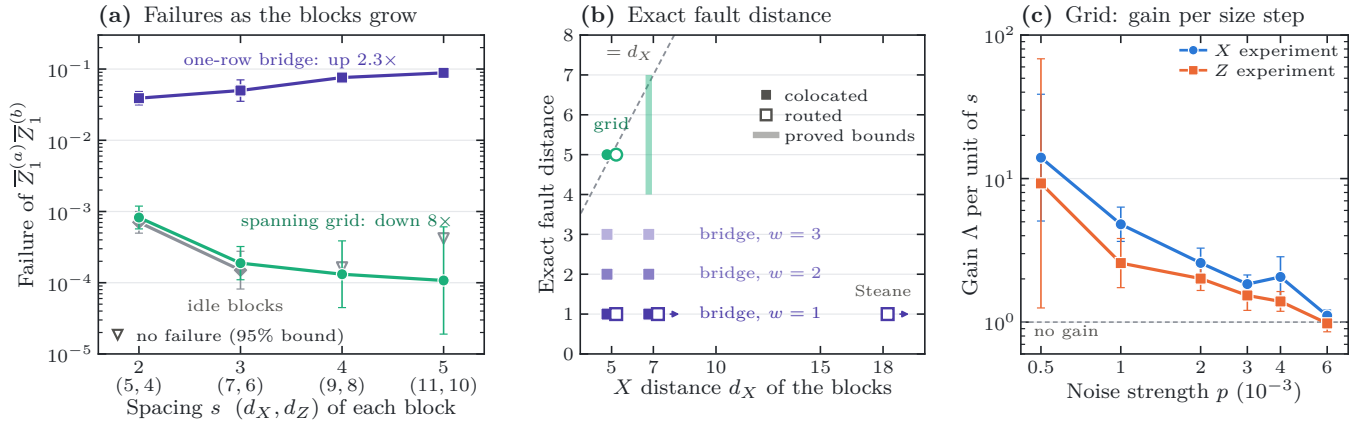


Figure 5: Two-block logical measurement on explicit layer-code blocks: the cut obstruction at scale, in simulation (the hardware counterpart is Fig. 6). (a) Failure of the measured pair $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ at $p = 10^{-3}$ as the block spacing s grows, through a one-row bridge (violet), a spanning grid (aqua), and for the blocks left idle (grey), with Wilson 95% intervals (open triangles: upper ends of zero counts). (b) Exact circuit fault distance against the X distance of the blocks, computed by CP-SAT. Filled: colocated ancillas at $s = 2, 3$; open: routed nearest-neighbor extraction at $s = 2$ with the $[[4, 2, 2]]$, band and Steane inputs (block X distances 5, at least 7 and at least 18, arrows); bar: proved lower bound and the weight of an undetectable pattern where the solver stopped. Through a bridge of w rows the distance stays at w; through the grid it equals d_X (dashed) where resolved. (c) Factor Λ by which each unit of spacing suppresses failures through the spanning grid, with 95% intervals.

on IBM), so the gain comes from the larger wall and not from discarding more samples. On IQM width three had no failure in 11,529 kept samples, but its difference from width two is not significant ($p = 0.071$, in a test fixed before the run): width two was already too rare to resolve. The IBM errors exceed the model, which assumed typical published error rates and no idle or crosstalk errors.

Parallel ancillas. Two parallel ancillas on a single bridge edge ($m_0 = 2$) give 0.53% instead of 2.4%, a 4.6-fold gain ($p = 5.1 \times 10^{-10}$), and 2.3% when the new two-edge check is ignored [Fig. 6(b,e)]. The first CNOT order we tried had circuit fault distance one: a fault on the vertex ancilla between its two bridge CNOTs put X on both parallel edges, which the two-edge check cannot see. Separating the two CNOTs restores distance two. This is the constant-factor loss from circuit faults that Sec. 2.4 allows for; a suitable CNOT order avoids it.

Repeated rounds and errors that survive a reset. With fresh ancillas in each round (IQM) the reported-bit error falls 20-fold, from 6.2% at $R = 1$ to 0.31% at $R = 2$, close to the model's 6.9% and 0.34% ($p = 2.4{\times}10^{-188}$, fixed in advance). On IBM, where the ancillas are measured and reset between rounds, it falls from 4.6% to 1.1% and 0.59% for $R = 1, 2, 3$ (both steps $p < 10^{-300}$, fixed in advance), but far less than the calibrated model predicts (0.18% and 0.0086% at $R = 2, 3$); at $R = 3$ the error is 68 times the prediction [Fig. 6(f)]. The cause is an ancilla error that survives the reset: a check ancilla reads 1 in two consecutive rounds, with no data error to explain it, in 1.15% of samples, 23 times the rate that independent errors give, and a second reset removes only 3% of these events ($p = 8.8 \times 10^{-6}$) [Fig. 6(g)]. Such a fault, most likely leakage out of the qubit subspace, corrupts the same outcome in every round. It lies outside the local-stochastic model of Sec. 2.4, which charges every round separately.

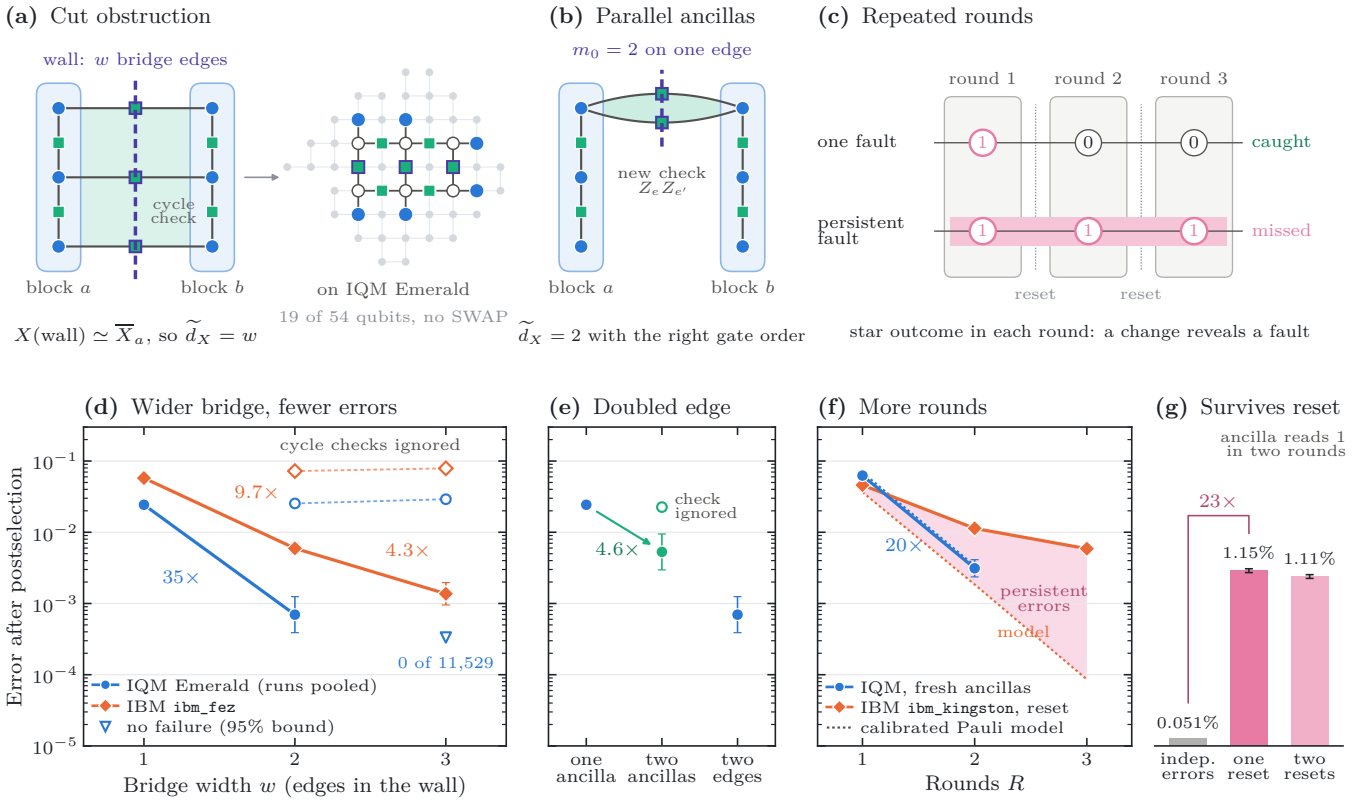


Figure 6: Hardware demonstrations of the three distance mechanisms. Top row: what was run. (a) Two three-qubit blocks (blue) joined by a bridge of w edges, each with an edge ancilla (aqua squares); X on the wall of bridge edges (violet) is equivalent to $\overline{X}_a$, so the deformed X distance and the exact circuit fault distance are both w. Shaded: the cycle checks that the extra bridge edges create. Right: the circuit for $w = 3$ as placed on IQM Emerald, with the couplers it uses. (b) Two parallel ancillas on one bridge edge ($m_0 = 2$) add the two-edge check $Z_e Z_{e'}$. (c) A wrong star outcome in one round is revealed by the change in the next round; a fault that persists across rounds repeats the same wrong outcome and is missed. Bottom row: what was measured, as the error after postselection on every detector, with Wilson 95% intervals (open triangle: upper end of a zero count). (d) IQM runs pooled; open symbols: the same samples with the cycle checks ignored. (e) IQM, one bridge edge with one or two ancillas, and two bridge edges; open symbol: the two-edge check ignored. (f) Reported-bit error against the number of rounds R, with the calibrated Pauli models (dotted); IQM uses fresh ancillas in every round, `ibm_kingston` measures and resets them. Shaded: the excess over the model (run A). (g) Rate at which a check ancilla on `ibm_kingston` (run B) reads 1 in two consecutive rounds with no data error to explain it, against the rate that independent errors give, with one or two resets between rounds.

4 Discussion

What the results mean. The answer to the research question is yes, under the stated noise model. Prefix-parity repair exposes logical operators in single-layer slabs of an explicit realization; grid gauging turns those slabs into protected measurements; and a composed detector schedule supports resource transfer and distillation in the same geometry. Because a layer-code distance is comparable to the area of a block face, every joint measurement must cross a constant fraction of that face. A thick grid does so and, with a constant ancilla multiplicity, keeps the full distance, whereas a thin measurement layer cannot. The simulations on layer-code blocks (Fig. 5) and the hardware runs (Fig. 6) show this contrast. The hardware runs reproduce at small scale the three facts that the distance argument combines: the smallest wall bounds the distance of a joint measurement, parallel ancillas widen every wall that crosses their edge, and repeated rounds protect the reported bit. For bounded-aspect logical regions the space bound separates an extensive Wf cost from the minimum $f^{3/2}$ cost of one encoded cell, and the port-packing result gives a second reading of the layer-code parameters: their cubic volume supports linearly many disjoint complete ports of each polarization.

Limitations. (i) The threshold is existential, and the decoder is minimum-weight, which is not efficient; a practical decoder for deformation histories is open. (ii) The constants are large: the storage advantage over surface codes appears only near $d \approx 1100$, and the distillation covered by the theorem needs injected errors below 2.3×10^{-3}. (iii) The fault-distance bounds are proved in the phenomenological model and transferred to circuits through the constant c_1; exact minimization checks them only on small instances. (iv) The hardware blocks are three-qubit codes, so the runs do not test layer codes, the grid expansion, the threshold or the resource scaling, which need far larger devices. The `ibm_fez` circuits were routed rather than SWAP-free, the first runs' tests were chosen after the data, and the pre-registered IQM test of width two against three was not met. (v) Ancilla errors that persist across rounds, most likely leakage, violate the local-stochastic assumption, so any implementation

Table 1: Hardware tests. Error after postselection on every detector; one-sided Fisher exact tests. P: test fixed before the run. Control: the same samples with the checks of the better interface ignored.

Device	Comparison	Error	Gain	p value
Cut obstruction				
IQM Emerald	$w = 1 \to 2$	$2.4\% \to 0.070\%$	$35\times$	$\leq 9.5 \times 10^{-19}$ (each run)
	$w = 2 \to 3$	$0.070\% \to 0$ of 11,529		0.071 (P, not met)
	control, $w = 2, 3$	2.2 to 3.3%		
IBM `ibm_fez`	$w = 1 \to 2$	$5.8\% \to 0.59\%$	$9.7\times$	1.7×10^{-278}
	$w = 2 \to 3$	$0.59\% \to 0.14\%$	$4.3\times$	1.1×10^{-16}
	control, $w = 2, 3$	7.2 to 7.9%		
Parallel ancillas				
IQM Emerald	one $\to$ two ancillas	$2.4\% \to 0.53\%$	$4.6\times$	5.1×10^{-10}
	control	2.3%		
Repeated rounds				
IQM Emerald	$R = 1 \to 2$, fresh ancillas	$6.2\% \to 0.31\%$	$20\times$	2.4×10^{-188} (P)
	model	$6.9\% \to 0.34\%$		
IBM `ibm_kingston`	$R = 1 \to 2 \to 3$, reset	$4.6\% \to 1.1\% \to 0.59\%$		$< 10^{-300}$ (P)
	model, $R = 2, 3$	0.18%, 0.0086%		
Ancilla errors that survive a reset				
IBM `ibm_kingston`	one $\to$ two resets	$1.15\% \to 1.11\%$		8.8×10^{-6} (P)
	independent errors	0.051%		

of the extraction schedule would need leakage reduction.

Future work. The next steps are a practical decoder for deformation histories, with changing checks, inherited frames and records, and joint measurements (the layer-code decoders of Ref. [39] are a natural starting point); routed circuit-level benchmarks on inputs that grow into the balanced regime; bounds on how many logical measurements can share the slab interfaces, compared with parallel code surgery [7]; factories with smaller constants; and an extension of the fault-path analysis to leakage and coherent noise. Random layer codes built from random CSS inputs [39] are a natural candidate for reducing the constants; the explicit realization applies to any CSS input, but the resource accounting would need their parameter scaling.

5 Conclusion

Three-dimensional layer codes can do more than store quantum information. Every pure logical operator of an explicit layer-code lattice has a representative in a single movable layer, and a local three-dimensional ancilla grid measures it without losing the optimal distance $\Theta(N^2)$, while every thin measurement interface provably fails. Built into a universal fault-tolerant architecture, these measurements keep the optimal storage cost, $O(Wf + f^{3/2})$ physical qubits for W logical sites. Simulations on explicit layer-code blocks and experiments on IQM and IBM processors confirm the distance mechanisms, and the experiments identify ancilla errors that survive a reset as the main assumption that present hardware does not meet.

Data and code availability

All code and data are in the public project repository [24]:

`github.com/HellWithCode0/layer-code-fault-tolerance`

It contains:

- `iris_paper/`: this paper, its source and figures;
- `supplement.pdf`: a technical report with the complete proofs, and `manuscript.pdf`, the companion journal manuscript;
- `layer_code_explicit.py` and `slab_gluing_certificate.py`: the explicit lattice, its check matrices and the slab certificates;
- `two_block_measurement.py`, `scaling_campaign/` and `numerical/`: the simulations, circuits, and every batch of samples with its seed;
- `min_weight_logical.py`: exact fault distances with CP-SAT;
- `hardware/`: the hardware circuits, job identifiers, raw outputs, calibration snapshots and analysis scripts; `hardware/hardware_summary.py` computes the hardware numbers of Sec. 3.8 and Table 1 from the raw outputs;
- `make_*_figure.py`: the figure scripts.

The repository README lists how to rerun each check.

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construction and the fault-distance argument be explained in plainer terms, and those parts were rewritten for that purpose. The correspondence did not include a detailed review of the proofs, and any errors are the author's own. The IQM experiments used the Open Quantum platform [30]. We acknowledge the use of IBM Quantum services for this work. The views expressed are those of the author, and do not reflect the official policy or position of IBM or the IBM Quantum team.

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